Corelogger Python Script (Version July 2026)

Purpose

The Corelogger Python Script can be used for multi-sensor core logger data preparation, analysis, and visualization.
The various measurements are recorded along the drill cores in specified intervals (see Fig. 1 for technical drawing of the GEOTEK MSCL).

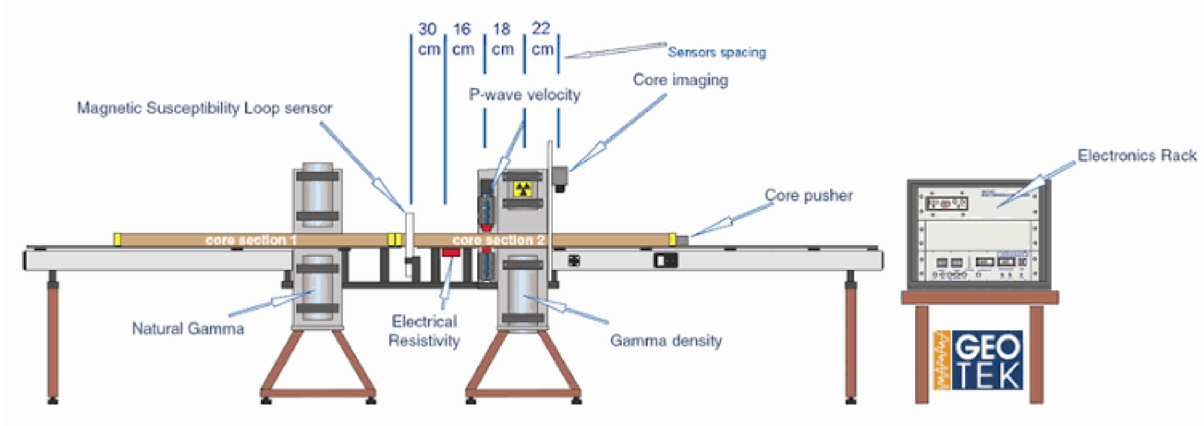


Fig. 1: GEOTEK Multi-Sensor Core Logger schematic (from Vatandoost, Adel & Fullagar, Peter & Walters, Steve & Kojovic, Toni. (2011). Towards petrophysical characterization of comminution behaviour.)

The Python script loads and formats the input data and visualizes it in 4 different plots:

a. Detailed visualisation for a specified core with accompanying core image (with the option of creating the plots for all cores at once),
b. Overview of the measurements for the entire depth range,
c. Crossplot of density and p-wave velocity &
d. Cluster-analysis crossplot of density and p-wave velocity

Requirements

This tool requires the following Python libraries:

- brokenaxes, matplotlib, numpy, pandas, scikit-learn, pillow
- built-in: io, math, pathlib, tkinter

The required libraries are listed again in the requirements.txt file. For easy installation execute the following line in the command prompt:

```
pip install -r requirements.txt
```

Lastly, a program to open and execute the .py script is required (e.g. Visual Studio Code).

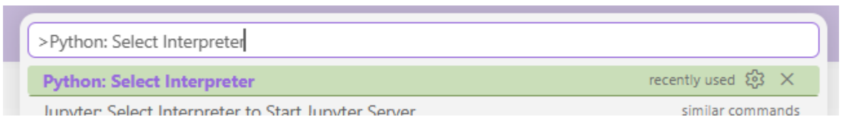
Step-by-Step guide

1. **Preparation**

- Cut and name core images (according to depth: e.g. “74-75.JPG”)
- Copy the images into a separate core image folder
- If core-section plots of all cores are to be made, create a designated output folder
- Automatic measurements will be extracted from .out files, which should be named after their core depth (e.g. “83-84.out”)
- Manual measurements need to be entered into an excel spreadsheet comparable to the following example:

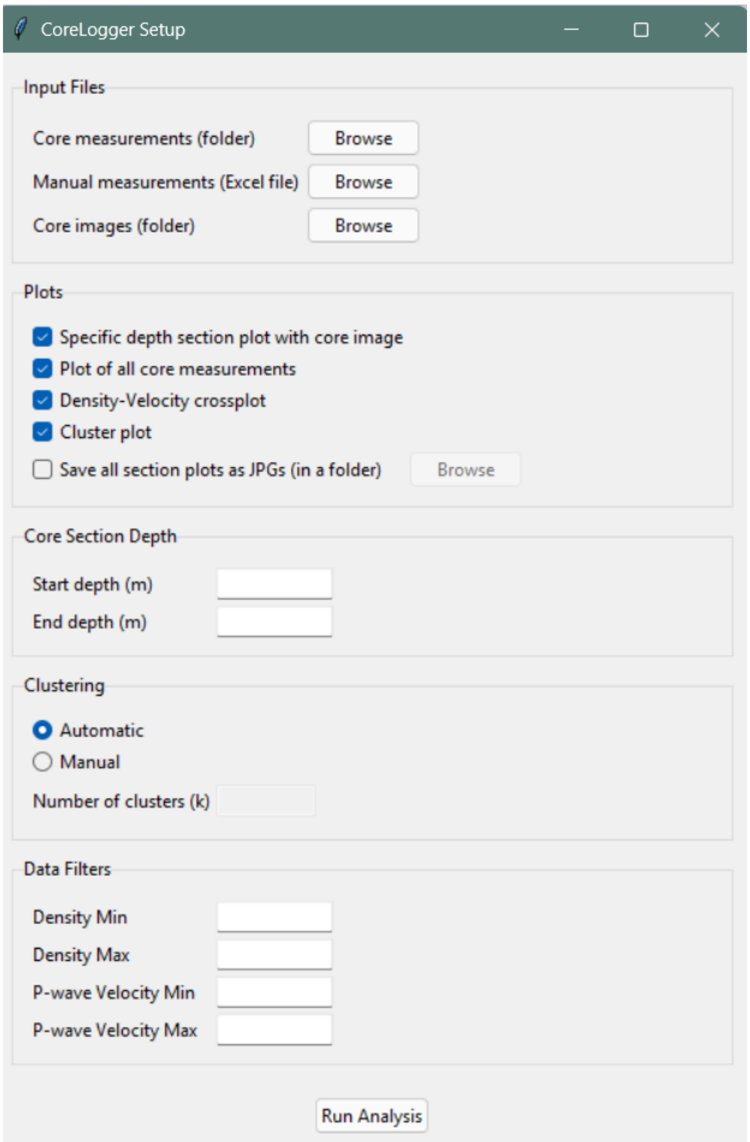
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Core name:		**DZA03Thonberg**												
2															
3	Core	Piece	Position	Diameter	Counts/s	Traveltime		Gamma-Kalibration			Rho_0	T_TOT			
4		nr	cm	cm	10 s	µs		A X2	B X	C		µs		Density	PwaveVel
5	**168-169**	26.01.2026						-0.0001	-0.0679	9.7131	2.602	14.46		g/cc	m/s
6		1	8	10.2	2407	39.2								2.716	4122.9
7		1	15	10.2	2418	38.9								2.709	4173.5
8		1	25	10.2	2448	39.45								2.692	4081.6
9		2	56	10.2	2458	39.9								2.686	4009.4
10		2	66	10.2	2500	38.8								2.663	4190.6
11		3	90	10.2	2487	37.7								2.670	4389.0
12		3	98	10.2	2431	37.2								2.702	4485.5
13															
14															
15															
16															
17	**93-94**	29.01.2026						-0.0001	-0.068	9.7119	2.602	14.41			
18		1	3	10.2	1957	30.85								3.000	6204.4
19		1	9	10.2	1917	31.5								3.029	5968.4
20		2	52	10.2	1946	30.7								3.008	6261.5
21		2	58	10.2	2010	30.95								2.962	6166.9
22		3	75	10.2	1920	30.5								3.027	6339.3
23		3	82	10.2	1939	30.75								3.013	6242.4
24		4	92	10.2	2015	30.9								2.959	6185.6
25															
26															

- Open Corelogger_Script.py with all necessary packages downloaded
  - In VS Code: You might need to turn off Restricted Mode or select to trust the file/folder to run the code
- Select Python Interpreter 3.13
  - In VS Code: Type “>Python: Select Interpreter” in the search bar (top of the window; see image below)



2. Running the code

- Run Code (VS Code: Top right corner play button ▸)
- The following Input Window will open:



The 3 Browse buttons will each lead to the Windows Explorer.
Choose the core measurements folder, then the excel file containing the manual measurements and lastly the core image folder.

Choose which plots you want to create.
You have the option of creating the core section plot for all cores at once.
If you check the box, you need to select an output folder (via Browse button) where the images will be saved.

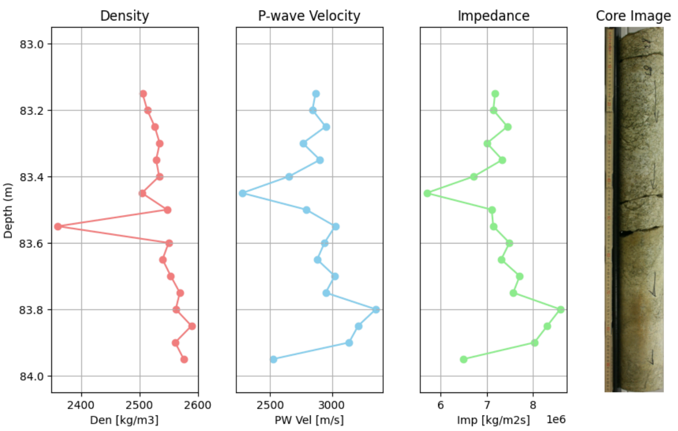
If you select the "Specific depth section" plot, choose a core (depth range) to plot its measurements and image (Plot 1).

Choose whether the number of clusters for the cluster analysis should be calculated automatically or put in manually.

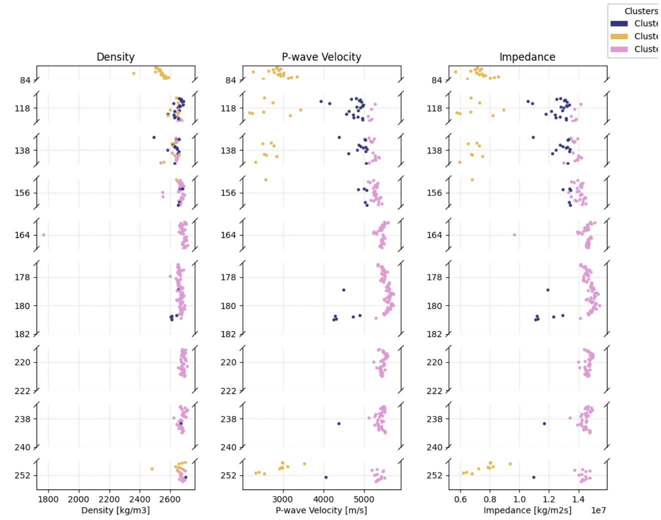
You can pecify minimum and maximum values for the density and p-wave velocity. Data outliers will not be shown on the plots.

Click the button Run Analysis.
After a few seconds the chosen plots will appear in separate windows (examples shown below).

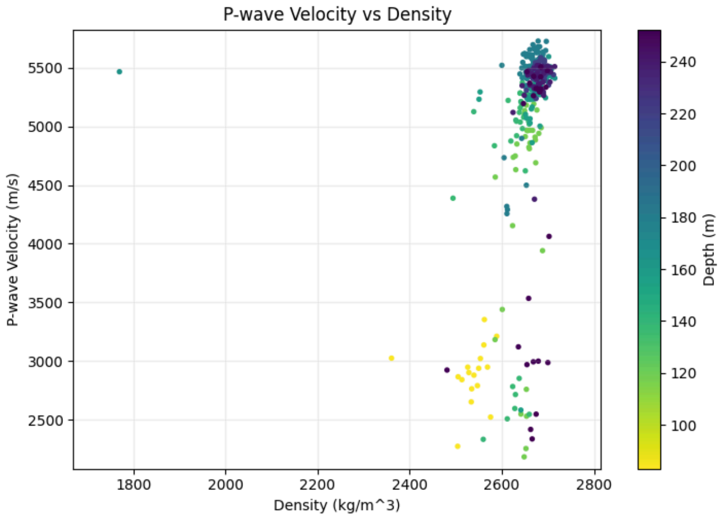
3. The Plots



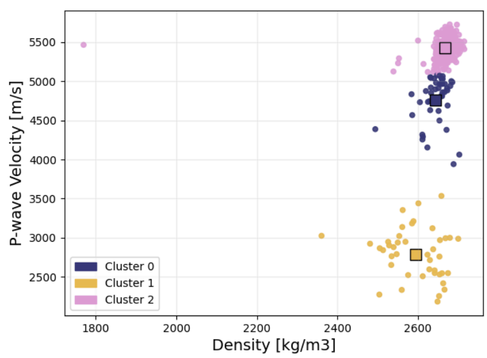
Plot 1 : Specific depth section with core image



Plot 2 : Entire depth range with cluster colors



Plot 3: Crossplot Density P-wave velocity with depth colormap



	Den min	Den max	Vp min	Vp max
C0	2493.9	2702.5	3940.9	5077.8
C1	2360.4	2699.9	2183.3	3534.1
C2	1769.6	2714.9	5102.0	5727.4

Plot 4: Cluster Crossplot with defining ranges

- The plots can be saved as JPG files to your computer by clicking this symbol on the gray bar at the bottom of the plot window: